

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ 2015MuZY

$J^\pi=0^+$ for ^{36}S ground state.

2015MuZY: A secondary beam ^{36}S was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a 846-mg/cm² ^9Be target at the coupled cyclotron facility at NSCL, MSU. The ^{36}S nuclei were selected using the A1900 separator with an intensity of 8.1×10^5 pps and a purity of 89.7%. The excited states of ^{35}S were populated by the one-neutron knockout reaction from the ^{36}S beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm² ^9Be secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt γ rays from the de-excitation of the ^{35}S residues were detected by seven modules of the GRETINA Ge array. Measured E_γ , I_γ , (^{35}S) γ -coin, $\gamma\gamma$ -coin, the inclusive knockout cross section for producing ^{35}S from ^{36}S , the fractional populations and parallel momentum distributions of 11 populated states in ^{35}S residues. Deduced levels, J , π , L-transfers, partial knockout cross sections and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for neutron removal and parallel momentum distributions using the eikonal model.

 ^{35}S Levels

The inclusive cross section $\sigma_{\text{inc}}^{\text{exp}}$ for producing ^{35}S from ^{36}S is measured to be 63.7 mb 6.

$E(\text{level})^\dagger$	$J^\pi^\#$	$L^\ddagger$	$C^2S^\text{@}$	Comments
0.0	$3/2^+$	2	3.2 7	$b_f=43.3\%$ 47, $\sigma_f^{\text{sp}}=13.1$ mb, $R_s=0.66$ 13.
1575 2	$1/2^+$	0	1.1 3	$b_f=18.6\%$ 31, $\sigma_f^{\text{sp}}=16.1$ mb, $R_s=0.64$ 13.
1991 2	$7/2^-$	3	<0.2	$b_f<2.9\%$, $\sigma_f^{\text{sp}}=12.7$ mb, $R_s=0.69$ 14.
2353 2	$3/2^-$	1	0.4 1	$b_f=5.3\%$ 13, $\sigma_f^{\text{sp}}=13.1$ mb, $R_s=0.70$ 14.
2720 2	$5/2^+$	2	0.6 2	$b_f=6.4\%$ 16, $\sigma_f^{\text{sp}}=11.4$ mb, $R_s=0.62$ 12.
2943 2	$(3/2)^+$	0	0.4 2	$b_f=5.7\%$ 16, $\sigma_f^{\text{sp}}=13.5$ mb, $R_s=0.61$ 12.
3425 2	$5/2^+$	2	0.5 1	$b_f=5.1\%$ 10, $\sigma_f^{\text{sp}}=11.0$ mb, $R_s=0.61$ 12.
3602 2	$(7/2^+)$	2	0.3 1	$b_f=3.0\%$ 7, $\sigma_f^{\text{sp}}=11.0$ mb, $R_s=0.60$ 12.
3821 3	$(9/2^-)$	3	0.3 1	$b_f=4.0\%$ 9, $\sigma_f^{\text{sp}}=11.8$ mb, $R_s=0.72$ 14.
3895 3	$(5/2^+)$	0	0.15 4	$b_f=2.3\%$ 5, $\sigma_f^{\text{sp}}=13.5$ mb, $R_s=0.72$ 14.
4028 2	$3/2^+, 5/2^+$	2	0.6 1	$b_f=6.4\%$ 10, $\sigma_f^{\text{sp}}=12.1$ mb, $R_s=0.60$ 12.

[†] From a least-squares fit to γ -ray energies.

[‡] 2015MuZY deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

[#] From the Adopted Levels. Inconsistencies exist between the 2015MuZY reported L values and J^π based on other datasets.

[@] The spectroscopic factor $C^2S=b_f\sigma_{\text{inc}}^{\text{exp}}/R_s\sigma_f^{\text{sp}}$, where b_f is the population fraction, $\sigma_{\text{inc}}^{\text{exp}}$ is the measured total inclusive cross section, R_s is the quenching factor from a systematics study 2014To14, and σ_f^{sp} is the eikonal model calculated single-particle cross section.

 $\gamma(^{35}\text{S})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π
778 2	2353	$3/2^-$	1575	$1/2^+$	2037 2	4028	$3/2^+, 5/2^+$	1991	$7/2^-$
1309 2	4028	$3/2^+, 5/2^+$	2720	$5/2^+$	2352 2	2353	$3/2^-$	0.0	$3/2^+$
1542 2	3895	$(5/2^+)$	2353	$3/2^-$	2720 2	2720	$5/2^+$	0.0	$3/2^+$
1576 2	1575	$1/2^+$	0.0	$3/2^+$	2943 2	2943	$(3/2)^+$	0.0	$3/2^+$
1830 2	3821	$(9/2^-)$	1991	$7/2^-$	3425 2	3425	$5/2^+$	0.0	$3/2^+$
1990 [‡] 2	1991	$7/2^-$	0.0	$3/2^+$	3602 2	3602	$(7/2^+)$	0.0	$3/2^+$

[†] Evaluators assume γ -ray energy uncertainties based on the statement in 2016Mu03 that γ -ray energy centroids were determined with an uncertainty of 2 keV.

[‡] Placement of transition in the level scheme is uncertain.

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Legend

Level Scheme-----► γ Decay (Uncertain)